Basic Information



Name: Zhang Zhanyuan
Gender: Male
Date of Birth: April 8, 2005
Phone: +86 185 3490 0536
Email: zhanyuan23@mails.tsinghua.edu.cn

Education

Tsinghua University, B.Eng. candidate

- **Admission**: August 2023 **Expected Graduation**: July 2027
- **Major**: Energy and Power Engineering, Department of Energy and Power Engineering
- **GPA**: 3.5 / 4.0 **Class Rank**: 15 / 28

Selected Core Coursework (Grade Point / Letter):

Core Course	GP	Core Course	GP
Engineering Thermodynamics	4.0 (A^-)	Fluid Mechanics	3.6 (B^+)
Heat Transfer	3.6 (B^+)	Engineering Mechanics A	3.6 (B^+)
Ordinary Differential Equations	4.0 (A^-)	Linear Algebra	3.6 (B^+)
Calculus A(2)	3.3 (B)	Probability & Statistics	3.6 (B^+)
Physics Laboratory	4.0 (A)	Engineering Drawing	4.0 (A)

Selected Major Coursework:

Course	GP	Course	GP
Big Data & Artificial Intelligence	4.0 (A^-)	Thermal Equipment Heat Transfer & Fluid Dynamics	4.0 (A)
Principles of Power Machinery & Engineering	3.6 (B^+)	Energy Physics	3.6 (B^+)
Fundamentals of Control Engineering	3.3 (B)	Energy Chemistry	3.0 (B^-)
Computer Programming Basics	3.3 (B)	Engineering Economics	4.0 (A^-)

Research Experience

Survey on Ceramic Matrix Composite (CMC) Turbine Blades for Aero-Engines

- **Supervisor**: Prof. Xinrong Su Oct 2023 – Sep 2025 | Completed
- Conducted a literature survey on the thermal resistance of ceramic matrix composites and their application in advanced aero-engine hot-end components; produced a survey report.

Biomimetic Hydrodynamic Suspension Rotor and Micropump Development

- **Supervisor**: Prof. Xianwu Luo Aug 2024 – Mar 2025 | Completed
- Investigated the multi-field coupling and rotor instability of a hydrodynamic-suspension micropump through literature review and biomimetic design; produced CAD drawings and SolidWorks models. Completed as a university "KeChuang YouCai" (Innovation Excellence) project.

Fluid–Structure Interaction of a Flexible NACA Airfoil

- **Supervisor**: Prof. Zhengwei Wang Sep 2025 – Jan 2026 | Completed
- Developed Python scripts to model a flexible NACA airfoil undergoing forced deformation under a directed incoming flow; implemented an iterative fluid–structure coupling scheme that converged and yielded conclusions consistent with qualitative expectations. Gained foundational experience in computational fluid dynamics and multi-physics coupling methods.

Battery Capacity Degradation Prediction

- **Supervisor**: Prof. Chao Liu Nov 2025 – Jan 2026 | Completed
- Independently completed data cleaning, machine-learning modeling, and optimization for battery capacity degradation; achieved $R^2 = 0.80$ with a multi-layer MLP after hyperparameter tuning. Learned the end-to-end pipeline of applying machine learning in the energy domain.

2D Ahmed Body External-Flow Simulation and Reduced-Order Surrogate Modeling

- **Supervisor**: Prof. Xingjian Wang — Feb 2026 – Jun 2026 | Completed
- Independently performed CFD modeling and simulation of the turbulent flow around a 2D Ahmed body to build a dataset; applied CAE-ROM (convolutional autoencoder-based reduced-order modeling) to reconstruct the flow field with high fidelity. Acquired hands-on experience in CFD workflows and reduced-order surrogate models.

Thermal Safety Analysis of a Simplified F1 Cockpit

- **Supervisor**: Prof. Hui Zhou — Apr 2026 – Jun 2026 | Completed
- Built a simplified F1 cockpit model retaining key heat-transfer processes and hot-end components; computed view factors with Monte Carlo ray tracing and validated the results with coupled conduction–convection heat transfer, reproducing conclusions consistent with the literature.

Enhancing Time-Series Forecasting via Future-Guided Learning

- **Supervisor**: Prof. Wenkai Liang — Apr 2026 – Present | Ongoing
- Reproduced the future-guided learning (FGL) method on the Mackey–Glass dataset and transferred it to a CSTR dataset, identifying the failure mode of FGL at excessively low MSE and the conditions under which it works; developed an adaptive distillation method to improve training. Deepened my understanding of deep learning methods and tools for time-series prediction.

Skills

- **Programming**: Python, C, basic MATLAB.
- **CFD & Numerical Methods**: CFD modeling and simulation; fluid–structure interaction; reduced-order surrogate modeling (CAE-ROM); Monte Carlo ray tracing; time-series forecasting.
- **Machine Learning**: Multi-layer perceptrons, autoencoders, future-guided learning, adaptive distillation, hyperparameter optimization.
- **Tools**: LaTeX, Git, SolidWorks, CAD.

Awards

- **Scholarships**: Tsinghua Friend – Dongfang Electric Scholarship (Volunteer & Public-Service Merit Scholarship, 2024); Tsinghua University University Scholarship (Volunteer & Public-Service Merit Scholarship, 2025).
- **Honor**: Tsinghua Five-Star "Zijing" Volunteer — May 2026

Activities & Volunteering

- **Leadership**: Chair, Publicity & Media Center of the Department (Feb 2026 – Present); Peer Mentor for new students (Aug 2024 – Jun 2025); Class Arts & Entertainment Officer (Sep 2024 – Aug 2025); Mentor at Ziqiang Academy (Aug 2025 – Present).
- **Volunteering**: Tsinghua Five-Star "Zijing" Volunteer with 242.5 accumulated hours, including the World Humanoid Robot Games (dance event, 42 h, Aug 2025), GT World Challenge (36 h, Oct 2025), and the "Siyuan" rural-education program (35 h, Jul 2025).
- **Sports**: Member of the departmental table-tennis team; competed twice in the "Ma Yuehan Cup" tournament.

Personal Statement

- Strong teamwork, organizational, and communication skills, with rich experience in volunteering, social practice, and student leadership.
- Solid foundation in mathematics and core energy-and-power courses, with ratings of B^+ or above in Fluid Mechanics, Heat Transfer, and Engineering Thermodynamics.
- Deeply interested in computational physics (fluid flow, combustion, and heat transfer) and its intersection with machine learning, especially reduced-order surrogate modeling and data-driven methods for flow analysis.
- Eager to pursue doctoral research on wind/tidal energy, turbomachinery flows, and machine-learning-assisted flow control.
- Optimistic and adaptable, with a steadily improving academic record during undergraduate study.